

The Ephemeris

June 2019

Volume 30 Number 02- The Official Publication of the San Jose Astronomical Association



June 2019 - September 2019 Events

Board & General Meetings

Saturday 6/15, 7/20, 8/17, 9/14
Board Meetings: 6 -7:30pm
General Meetings: 7:30-9:30pm

Fix-It Day (2-4pm)

Sunday 6/2, 7/7, 8/4, 9/8

Solar Sunday (1:30-3:30pm)

Sunday 6/2, 7/7, 8/4, 9/8

Intro to the Night Sky Class & Houge Park 1Q In-Town Star Party

Friday 6/7, 7/12, 8/9, 9/6

Houge Park 3Q In-Town Star Party

Friday 6/21, 7/26, 8/23, 9/20

RCDO Starry Nights Star Party

Saturday 6/22, 7/27, 8/24, 9/21

Imaging SIG Mtg

Tuesday 6/18, 7/16, 8/20, 9/17

Astro Imaging Clinics

(at Little Uvas OSP)

Saturday 6/29, 7/27, 8/24

Astro Imaging Workshops

(at Little Uvas OSP)

Saturday 6/1, 9/28

Quick STARt

By appointment

Binocular Star Gazing

Saturday 6/22, 7/20, 8/24

Please refer to the SJAA Website home page (www.sjaa.net) and click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancellation due to weather.

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Photo by SJAA Member: Glenn Newell
Feb 20, 2019
Location: Union City, CA

President's Note

By Swami Nigam

Dear SJAA Members,

The first few months of this year have brought a lot of welcome rain to our area. But it hasn't been a very astronomy friendly weather. With warmer weather and clear skies just around the corner, I can't wait to get my astronomical gear out into the open again and gather some much needed star light. I hope you are looking forward to the summer months as much as I am.

As you already know, the SJAA officer elections were held at the March board meeting. It is my honor to be elected to serve as the President for another year. Also, Rob Jaworski was elected to the position of the Treasurer for another term. And Sandy Mohan was elected to fill the position of the Secretary vacated by Rob Chapman.

I would like to thank Rob Chapman and recognize many years of dedicated service to the club as the Secretary. Not only did Rob fulfill all the duties of the Secretary, but he also went beyond the ask and was instrumental in setting up and managing the sjaa.net email and account infrastructure. Rob, your contribution is greatly appreciated. I hope you continue to stay involved with the club to whatever extent you possibly can.

I would also like to welcome Sandy Mohan to the new position of the Secretary. Most of you already know Sandy as the Content Editor for The Ephemeris for many years. After stepping down from that position last year, Sandy joined the board as a Director a few months back, and now has taken over the responsibilities of the Secretary. Sandy, welcome to your new role.

The position of the Vice-President continues to remain open. The club operations will benefit tremendously if we can fill this position quickly. If any of you are interested in helping the club operations by taking on this role, please reach out to me. Note, being on the Board of Directors

is NOT a prerequisite to taking on an officer role. All club members in good standing are eligible to apply for this position.

In addition to the officer positions, the club also has many "Leader" positions, including some that are being newly created, that are currently open. I will be announcing these open positions shortly on our website. I will end my note here with a sincere appeal to all club members that if you are able to spare a few hours a month to help the club in its operations, please do step forward and let us know.

Please email me with any ideas and suggestions at president@sjaa.net.

With best regards,

Swami Nigam

Spring Swap Meet Recap



The late Spring SJAA Swap Meet was a fun event for all of the attendees. The facility was near capacity. The rain probably kept a good number of folks from attending but for those buyers who did show up there were plenty of sellers. I saw a number of our members score some really good deals. The club only netted \$540 from the swap meet but did pick up \$975 from the donation sales.

The next Swap Meet will be held on Sunday October 13th.

Black Hole Image Makes History

Credit: NASA JPL



Scientists have obtained the first image of a black hole, using Event Horizon Telescope observations of the center of the galaxy M87. The image shows a bright ring formed as light bends in the intense gravity around a black hole that is 6.5 billion times more massive than the Sun. Credit: Event Horizon Telescope Collaboration

A black hole and its shadow have been captured in an image for the first time, a historic feat by an international network of radio telescopes called the Event Horizon Telescope (EHT). EHT is an international collaboration whose support in the U.S. includes the National Science Foundation.

A black hole is an extremely dense object from which no light can escape. Anything that comes within a black hole's "event horizon," its point of no return, will be consumed, never to re-emerge, because of the black hole's unimaginably strong gravity. By its very nature, a black hole cannot be seen, but the hot disk of material that encircles it shines bright. Against a bright backdrop, such as this disk, a black hole appears to cast a shadow.

The stunning new image shows the shadow of the supermassive black hole in the center of Messier 87 (M87), an elliptical galaxy some 55 million light-years from Earth. This black hole is 6.5 billion times the mass of the Sun. Catching its shadow involved eight ground-based radio telescopes around the globe, operating together as if they were one telescope the size of our entire planet.

"This is an amazing accomplishment by the EHT team," said Paul Hertz, director of the astrophysics division at NASA Headquarters in Washington. "Years ago, we thought we would have to build a very large space telescope to image a black hole. By getting radio telescopes around the world to work in concert like one instrument, the EHT team achieved this, decades ahead of time."

To complement the EHT findings, several NASA spacecraft were part of a large effort, coordinated by the EHT's Multiwavelength Working Group, to observe the black hole using different wavelengths of light. As part of this effort, NASA's Chandra X-ray Observatory, Nuclear Spectroscopic Telescope Array (NuSTAR) and Neil Gehrels Swift Observatory space telescope missions, all attuned to different varieties of X-ray light, turned their gaze to the M87 black hole around the same time as the EHT in April 2017. NASA's Fermi Gamma-ray Space Telescope was also watching for changes in gamma-ray light from M87 during the EHT observations. If EHT observed changes in the structure of the black hole's environment, data from these missions and other telescopes could be used to help figure out what was going on.

While NASA observations did not directly trace out the historic image, astronomers used data from NASA's Chandra and NuSTAR satellites to measure the X-ray brightness of M87's jet. Scientists used this information to compare their models of the jet and disk around the black hole with the EHT observations. Other insights may come as researchers continue to pore over these data.

There are many remaining questions about black holes that the coordinated NASA observations may help answer. Mysteries linger about why particles get such a huge energy boost around black holes, forming dramatic jets that surge away from the poles of black holes at nearly the speed of light. When material falls into the black hole, where does the energy go?

"X-rays help us connect what's happening to the particles near the event horizon with what we can measure with our telescopes," said Joey Neilsen, an astronomer at Villanova University in Pennsylvania, who led the Chandra and NuSTAR analysis on behalf of the EHT's Multiwavelength Working Group.

NASA space telescopes have previously studied a jet extending more than 1,000 light-years away from the center of M87. The jet is made of particles traveling near the speed of light, shooting out at high energies from close to the event horizon. The EHT was designed in part to study the origin of this jet and others like it. A blob of matter in the jet called HST-1, discovered by Hubble astronomers in 1999, has undergone a mysterious cycle of brightening and dimming.

NASA's LRO sheds light on Lunar Water Movement

Credit: NASA JPL



Using an instrument aboard NASA's Lunar Reconnaissance Orbiter (LRO), scientists have observed water molecules moving around the dayside of the Moon.

A paper published in *Geophysical Research Letters* describes how Lyman Alpha Mapping Project (LAMP) measurements of the sparse layer of molecules temporarily stuck to the surface helped characterize lunar hydration changes over the course of a day.

Up until the last decade or so, scientists thought the Moon was arid, with any water existing mainly as pockets of ice in permanently shaded craters near the poles. More recently, scientists have identified surface water in sparse populations of molecules bound to the lunar soil, or regolith. The amount and locations vary based on the time of day. This water is more common at higher latitudes and tends to hop around as the surface heats up.

"This is an important new result about lunar water, a hot topic as our nation's space program returns to a focus on lunar exploration," said Dr. Kurt Retherford, the principal investigator of the LAMP instrument from Southwest Research Institute in San Antonio, Texas. "We recently converted the LAMP's light collection mode to measure reflected signals on the lunar dayside with more precision, allowing us to track more accurately where the water is and how much is present."

Water molecules remain tightly bound to the regolith until surface temperatures peak near lunar noon. Then, molecules thermally desorb and can bounce to a nearby location that is cold enough for the molecule to stick or populate the Moon's extremely tenuous atmosphere or exosphere, until temperatures drop and the molecules return to the surface. SwRI's Dr. Michael Poston, now a research scientist on the LAMP team, had previously conducted extensive experiments with water and lunar samples collected by the Apollo missions. This research revealed the amount of energy needed to remove water molecules from lunar materials, helping scientists understand how water is bound to surface materials.

"Lunar hydration is tricky to measure from orbit, due to the complex way that light reflects off of the lunar surface,"

Poston said. "Previous research reported quantities of hopping water molecules that were too large to explain with known physical processes. I'm excited about these latest results because the amount of water interpreted here is consistent with what lab measurements indicate is possible.

Scientists have hypothesized that hydrogen ions in the solar wind may be the source of most of the Moon's surface water. With that in mind, when the Moon passes behind the Earth and is shielded from the solar wind, the "water spigot" should essentially turn off. However, the water observed by LAMP does not decrease when the Moon is shielded by the Earth and the region influenced by its magnetic field, suggesting water builds up over time, rather than "raining" down directly from the solar wind.

"These results aid in understanding the lunar water cycle and will ultimately help us learn about accessibility of water that can be used by humans in future missions to the Moon," said Amanda Hendrix, a senior scientist at the Planetary Science Institute and lead author of the paper. "Lunar water can potentially be used by humans to make fuel or to use for radiation shielding or thermal management; if these materials do not need to be launched from Earth, that makes these future missions more affordable."

"This result is an important step in advancing the water story on the Moon and is a result of years of accumulated data from the LRO mission," said John Keller, LRO deputy project scientist from NASA's Goddard Space Flight Center in Greenbelt, Maryland.

Goddard manages the LRO mission for the Science Mission Directorate at NASA Headquarters in Washington, D.C. Funding for the research came from LRO, and the team received additional support from a NASA Solar System Exploration Research Virtual Institute (SSERVI) cooperative agreement.

NASA is leading a sustainable return to the Moon with commercial and international partners to expand human presence in space and bring back new knowledge and opportunities.

Rosette Nebula

Rick McIlmoil



Date: Jan 22, 2019

Location: Backyard, Newark, CA

Telescope: Astro-Tech 8" RC

Camera: Nikon D5100

Mount: Celestron CGEM Equatorial Mount

Software: Stellarium Developers Stellarium 0.16.1, Sequence Generator Pro, PHD2, Backyard Nikon, Stellarium, Stellarium Scope, AstroTortilla 0.7, DSS, Adobe Photoshop Photoshop CC (Cloud)



Dates: Mar 16, 2019, Mar 24. 2019

Location: San Jose, CA

Telescope: Takahashi FS 60CB

Camera: Orion Star Shoot G3 Mono

Mount: Orion Sirius EQ-G

Software: Photoshop CC 2018

SJAA Member Astrophoto Gallery



Date: Sept 8, 2018; Jan 24, 2019

Location: Lake San Antonio, Bradley, California; Pinnacles National Park, Hollister, California

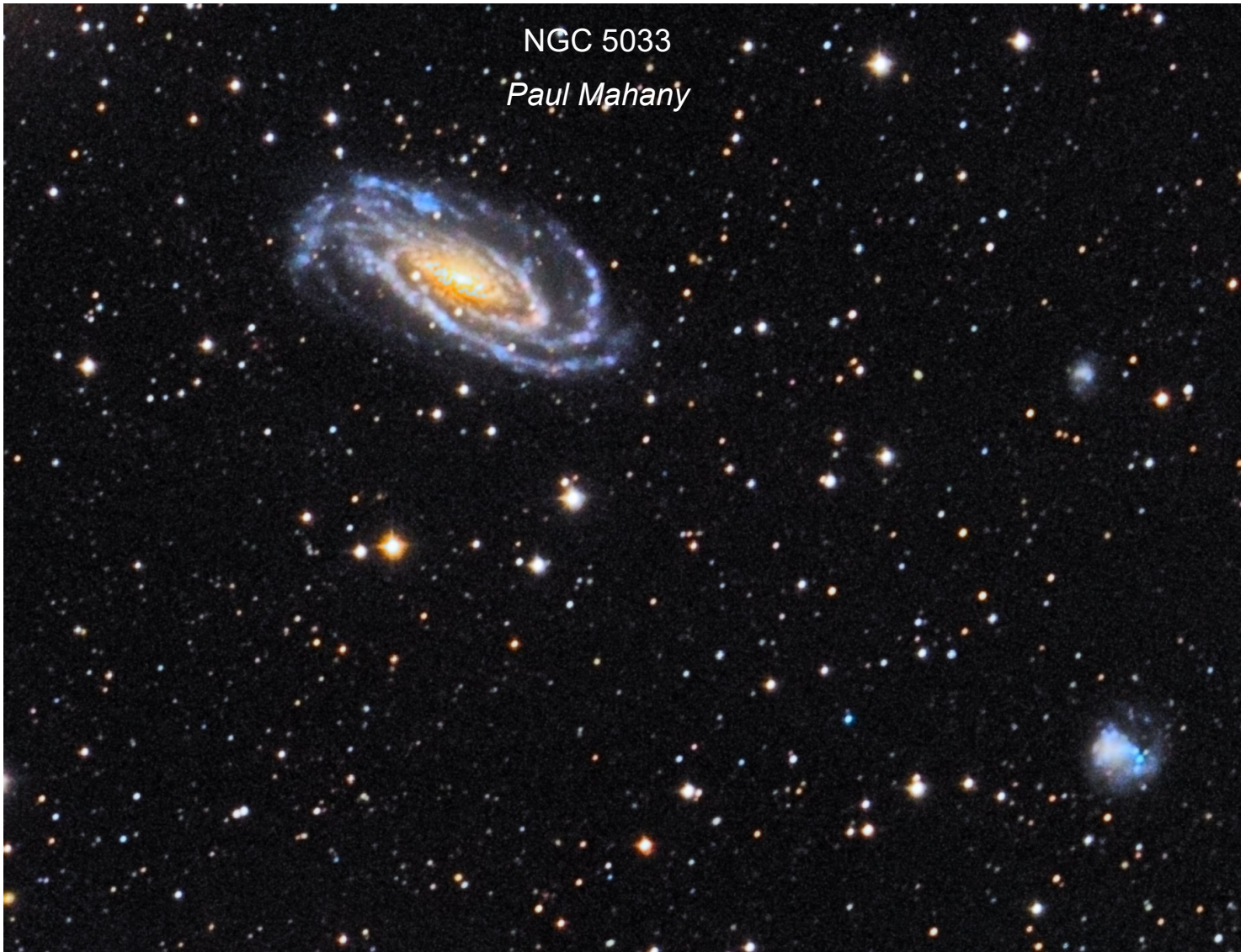
Telescope: Explore Scientific 102mm ED APO triplet

Camera: Cannon T3i MODDED

Mount: Orion Atlas Pro AZ/EQ-G

Software: Pleiades Astrophoto PinInsight

SJAA Member Astrophoto Gallery



Date: April 10 2019

Location: San Jose, Ca

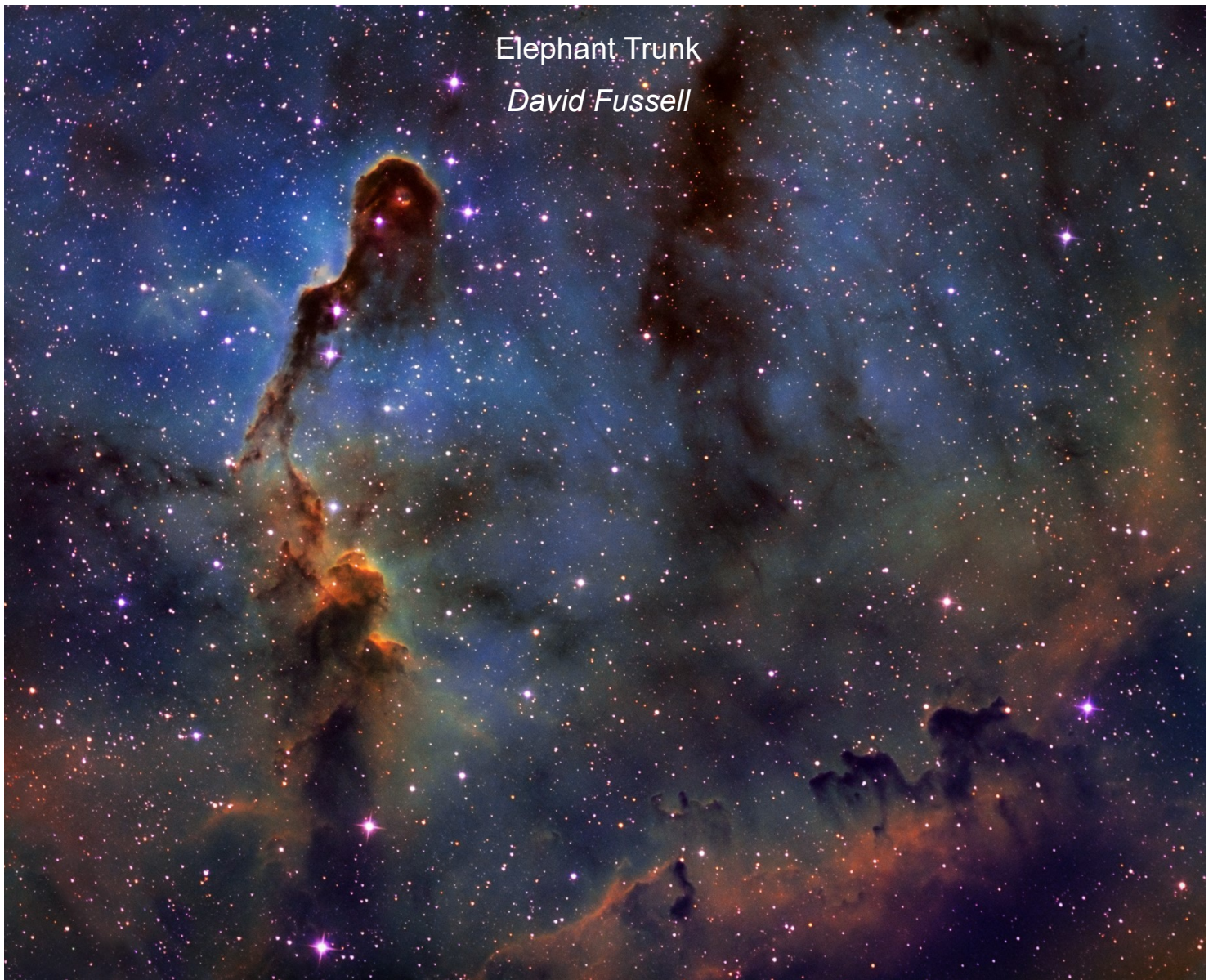
Telescope: Celestron RASA

Camera: Orion StarShoot Autoguider

Mount: Software Bisque Paramount MX+

Software: TheSkyX Pro, PixInsight1.8, SkyTools, BYEOS, PHD Guiding 2 PHD 2 Guiding, StarTools

SJAA Member Astrophoto Gallery



Date: Dec. 20, 2017, Dec. 21, 2017, Dec. 22, 2017

Location: Campbell, CA

Telescope: TS Optics 8" f/4 ONTC Newton

Camera: ZWO Optical ZWO ASI1600MM Cool

Mount: Orion Atlas Pro AZ/EQ-G

Software: Pleiades Astrophoto PixInsight 1.8 Ripley



Kid Spot Jokes:

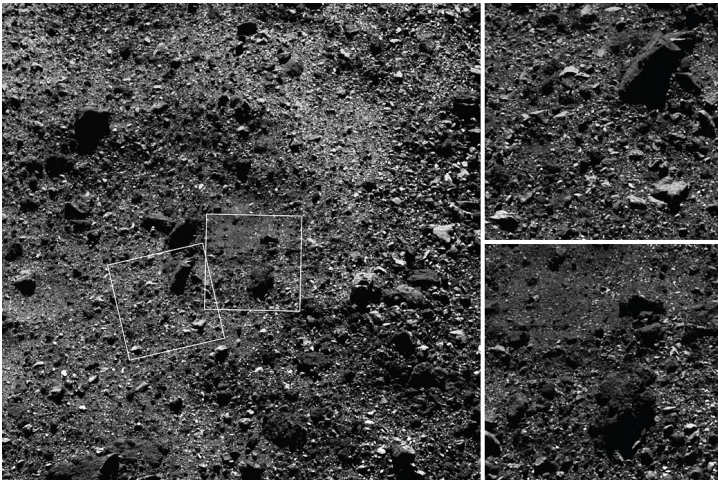
- ♦ **What is a pirate's favorite planet?**
Marrrrs.
- ♦ **What do you call someone who doesn't eat animal products and loves astronomy?**
Vega-n
- ♦ **What star do you look at when it is very cold outside?**
Alde-brrrr-an

A Region of Bennu's Northern Hemisphere Close Up

This trio of images acquired by NASA's OSIRIS-REx spacecraft shows a wide shot and two close-ups of a region in asteroid Bennu's northern hemisphere. The wide-angle image (left), obtained by the spacecraft's MapCam camera, shows a 590-foot (180-meter) wide area with many rocks, including some large boulders, and a "pond" of regolith that is mostly devoid of large rocks. The two

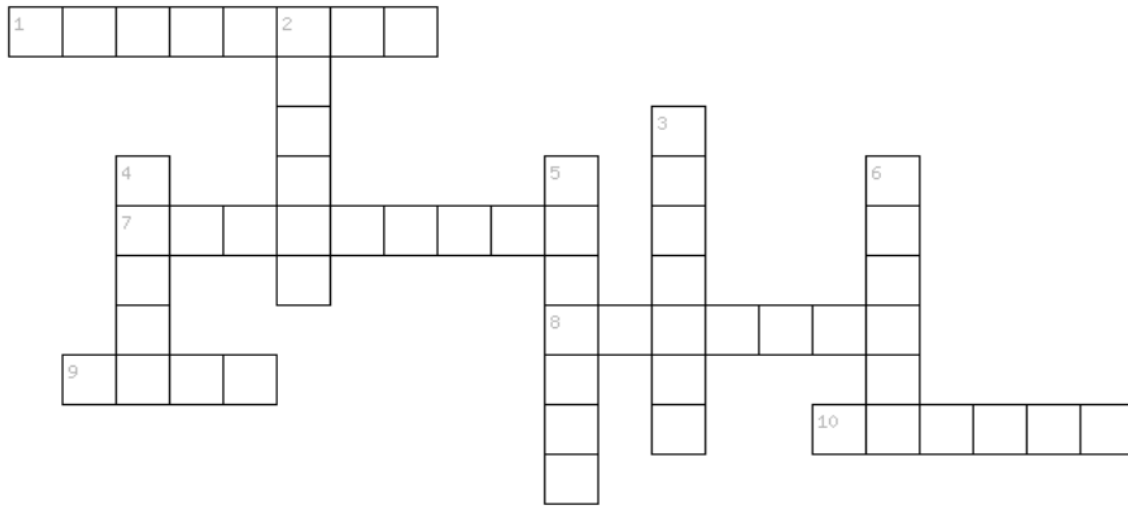
ters) across and the boulder depicted is approximately the same size as a humpback whale.

The images were taken on February 25 while the spacecraft was in orbit around Bennu, approximately 1.1 miles (1.8 km) from the asteroid's surface. The observation plan for this day provided for one MapCam and two PolyCam images every 10 minutes, allowing for this combination of context and detail of Bennu's surface.



closer images, obtained by the high-resolution PolyCam camera, show details of areas in the MapCam image, specifically a 50-foot (15 meter) boulder (top) and the regolith pond (bottom). The PolyCam frames are 101 feet (31 me-

Kids Spot Crossword Puzzle



Down

2. Icy moon of Jupiter
3. Lakes on Titan are made of this
4. Scientist who proposed paradox asking why aliens are not here
5. Desert which houses many space telescopes
6. 7 sisters stars

Across

1. Largest moon in the solar system
7. A planet orbiting another star
8. X-Ray observatory
9. Observatory which detected gravitational waves
10. Scientist who discovered that space is expanding

Answers

Down:

2. Europa
3. Methane
4. Fermi
5. Atacama
6. Subaru

Across:

1. Ganymede
7. Exoplanet
8. Chandra
9. LIGO
10. Hubble

From the Board of Directors

Announcements

- ◇ Sandy Mohan has been elected as Secretary position replacing Rob Chapman.
- ◇ SJAA has been assigned July 26 and 27 for the Stars Over Yosemite party.
- ◇ SJAA is working with the USS Hornet museum and the Tech Museum of Innovation in San Jose to collaborate in the 50th anniversary celebration
- ◇ SJAA Library is operational and available to all members.

Board Meeting Excerpts

March 23, 2019

In attendance

Swami, Rob C, Sukhada, Glenn, Ken, Vini, Sandy, Wolf
Excused Absences: Rob J
Guest: Alexandra

Officer Elections

Swami Nigam and Rob Jaworski re-elected to the positions of President and Treasurer, respectively. Rob Chapman has stepped down from the position of Secretary. Sandy Mohan was elected as the new Secretary. Vice-President position is still open.

Scout-a-Rama on May 11

SJAA will setup a booth at the Scout-o-Rama boy scout event on May 11 at History Park in San Jose.

Stars Over Yosemite Party

SJAA has been assigned July 26 and 27 for the Stars Over Yosemite party. Ken will coordinate the event.

Meetup Group

Description for our Meetup group SJ Astronomy to be updated to reflect that joining the meetup is not the same as becoming a club member.

April 20, 2019

In attendance

Sukhada, Wolf, Vini, Rob J, Rob C
Excused Absences: Swami, Sandy, Glenn, Ken
Guest: None

Outreach Request Policy

Sandy is working on an official SJAA policy for outreach requests received by SJAA from various sources.

USS Hornet museum and the Tech Museum

SJAA is working with the USS Hornet museum in San Francisco and the Tech Museum of Innovation in San Jose to collaborate/participate in the 50th anniversary celebration of the lunar landing.

Preschool lights

Vini and Leslie are interacting with the Cambrian school district to address the presence of bright lights on their buildings adjacent to the Houge Park parking lot, which interferes with providing dark conditions for star gazing during ITSP/

Library

SJAA Library is operational and available to all members.

May 18, 2019

In attendance

Swami, Wolf, Vini, Sukhada, Glenn, Sandy
Excused Absences: Ken, RobJ, RobC
Guest: Marianne

QuickSTART Program

QuickSTART program is currently on hold, pending some restructuring and additional volunteer help.

Scout-o-Rama event

Scout-o-Rama boy scout event went well. Lots of visitors during the 6 hours of booth time. Visitors were quite awed with viewing the Sun.

Kelley Park

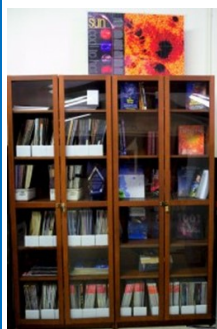
Interactions with the Kelley Park and San Jose rangers has been very intermittent regarding the 2 dates they have on their calendar for astronomy related event. Swami reached out to them again to firm up the plans, if any.

Pending outreach requests

Pending outreach requests from Montalvo Arts Center in Saratoga, and Rotary Club, San Jose. Decision pending

on Montalvo. Will not participate in Rotary Club as it's not an educational event.

SJAA Library



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments

to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick START Program

The Quick START Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick START program is undergoing a redesign to make it more accessible to SJAA members, so signups are currently on hold.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by David Grover, each with a similar format, with only the content changing to reflect what's currently in the night

sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met. If you are an SJAA member and an experienced observer or have been through the SJAA Quick START program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Bruce Braunstein, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astrophotography, to have a chance of seeing what it is all about.

Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's (Program Coordinator) for

additional information.

<http://www.sjaa.net/programs/school-star-party/>

SJAA Contacts

President/Dir:	Swami Nigam
Vice President:	Vacant
Treasurer/Dir:	Rob Jaworski
Secretary/Dir:	Sandy Mohan
Director:	Rob Chapman
Director:	Vini Carter
Director:	Ken Miura
Director:	Glenn Newell
Director:	Wolf Witt
Director:	Sukhada Palav
Ephemeris Newsletter -	
Content Editor:	Vikas Kache
Prod. Editor:	Emi Nikolov
Binocular Stargazing:	Ed Wong
Fix-it Program:	Vini Carter
Hands on Imaging:	Glenn Newell
Imaging SIG:	Bruce Braunstein
Intro to the Night Sky:	David Grover
Library:	Jai Purandare
Loaner Program:	Muditha Kanchana
Memberships:	Kaushal Gangakhedkar
Publicity:	Rob Jaworski
Quick START	Dave Ittner
Donations:	Dave Ittner*
Solar:	Wolf Witt
Starry Nights:	Carl Wong
School Events:	Cat Cvengros
Refreshments:	Marianne Damon
Social Media Manager:	Jessica Johnson
Speakers:	Sukhada Palav
Webmaster:	Satish Vellanki
E-mails:	http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

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San Jose Astronomical Association Annual Membership Form

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Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

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